

Mise

Household Cooking Assistant

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Mahfood

Less waste. Better meals. Together.

The Problem

- Households waste food because they don't know what they have
 - People buy duplicates, forget expiring items, and can't coordinate cooking
 - There's no tool that connects your pantry to what you can actually cook today
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Mise — What It Does

- ▶ Scan or manually add items to a shared household pantry
 - ▶ Get recipe suggestions ranked by what you already have
 - ▶ Full matches: cook now. Partial matches: see what's missing
 - ▶ Shared real-time shopping list across all household members
 - ▶ Works offline. Syncs automatically when back online.
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Technical Architecture

- ▶ **Flutter** — cross-platform mobile
 - ▶ **BLoC** — state management (PantryBloc, ShoppingBloc, CookBloc)
 - ▶ **Drift (local SQLite)** — offline-first data layer
 - ▶ **Firebase Auth** — email/password authentication
 - ▶ **Cloud Firestore** — real-time sync across devices
 - ▶ **Spoonacular API** — recipe suggestions by ingredient
 - ▶ **OpenFoodFacts API** — barcode product lookup
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State Management with BLoC

- ▶ **3 BLoCs:** PantryBloc, ShoppingBloc, CookBloc
 - ▶ **PantryBloc events:** PantryStarted, PantryItemAdded, PantryItemRemoved
 - ▶ **PantryBloc states:** PantryInitial, PantryLoading, PantryLoaded, PantryError
 - ▶ **UI never talks to data directly:** only sends events, reacts to states
 - ▶ **CookBloc:** triggered automatically by a BlocListener on PantryBloc — pantry changes instantly refresh recipe suggestions
 - ▶ **ShoppingBloc:** checking off an item removes it from shopping AND adds it to pantry automatically
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Offline-First Data with Drift

- ▶ All data is written to local SQLite (Drift) first — app works with no internet
 - ▶ Every item has an isSynced flag (default: false)
 - ▶ When online: item is pushed to Firestore → isSynced set to true
 - ▶ When offline: item stays with isSynced = false
 - ▶ On reconnect: syncPending() finds all isSynced=false rows and retries
 - ▶ Items arriving from Firestore are inserted with isSynced = true
 - ▶ Tables: Users, Households, PantryItems, ShoppingListItems
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Authentication & Cross-Device Data

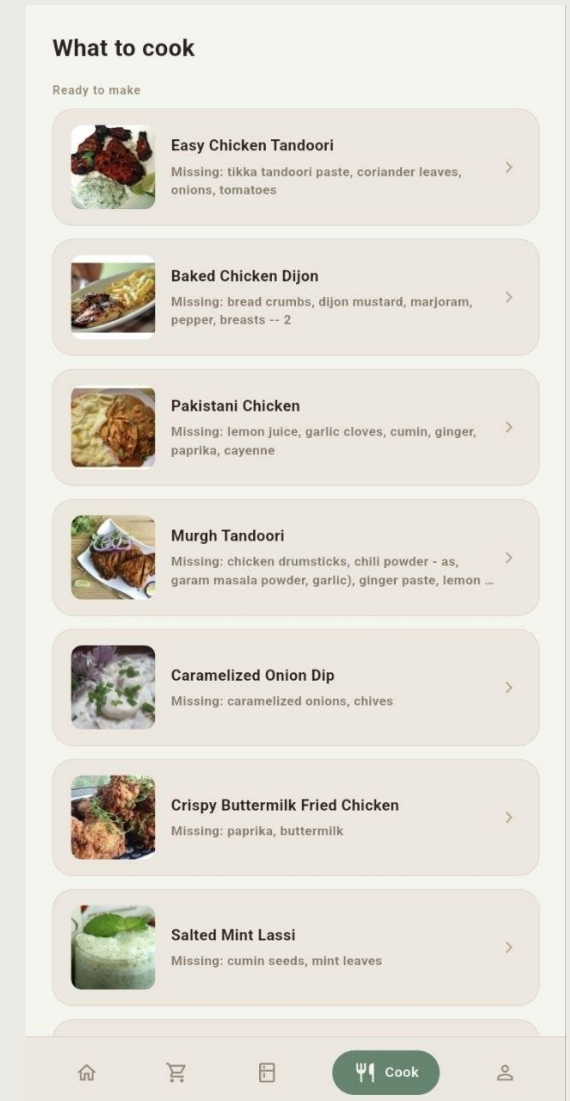
- ▶ Firebase Auth: email/password sign up and sign in
 - ▶ On login: MainShell loads householdId from Firestore users/{uid}
 - ▶ All pantry and shopping data is scoped to households/{householdId}/
 - ▶ Logging in on a new device automatically loads all household data from Firestore
 - ▶ Households are created or joined via invite code
 - ▶ Auth routing: no auth → AuthScreen | auth + no household → HouseholdSetupScreen | auth + household → main app
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Real-Time Collaboration

- ▶ Firestore uses persistent WebSocket connections for real-time listeners
 - ▶ `PantryRepository.listenToFirestore()` streams all pantry changes across devices
 - ▶ `ShoppingRepository.listenToFirestore()` handles added, modified, and removed events
 - ▶ Full flow: User A adds "tomato" → Drift insert + Firestore push → User B's listener fires → Drift upsert → `PantryBloc` rebuilds UI → `CookBloc` auto-fetches new recipes
 - ▶ `HomeScreen` shows a live shopping list preview that updates instantly
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Smart Recipe Suggestions

- ▶ **API:** Spoonacular /recipes/findByIngredients
- ▶ **Two-call strategy:** ranking=1 (maximize used ingredients) + ranking=2 (minimize missed)
- ▶ **ignorePantry=true:** to avoid false matches on staples like salt and water
- ▶ **Full matches:** missedIngredientCount = 0 → "Ready to make"
- ▶ **Partial matches:** missedIngredientCount > 0 → "Almost there" with missing items listed
- ▶ **Tap any recipe:** detail sheet with ingredients and cooking instructions (HTML rendered)

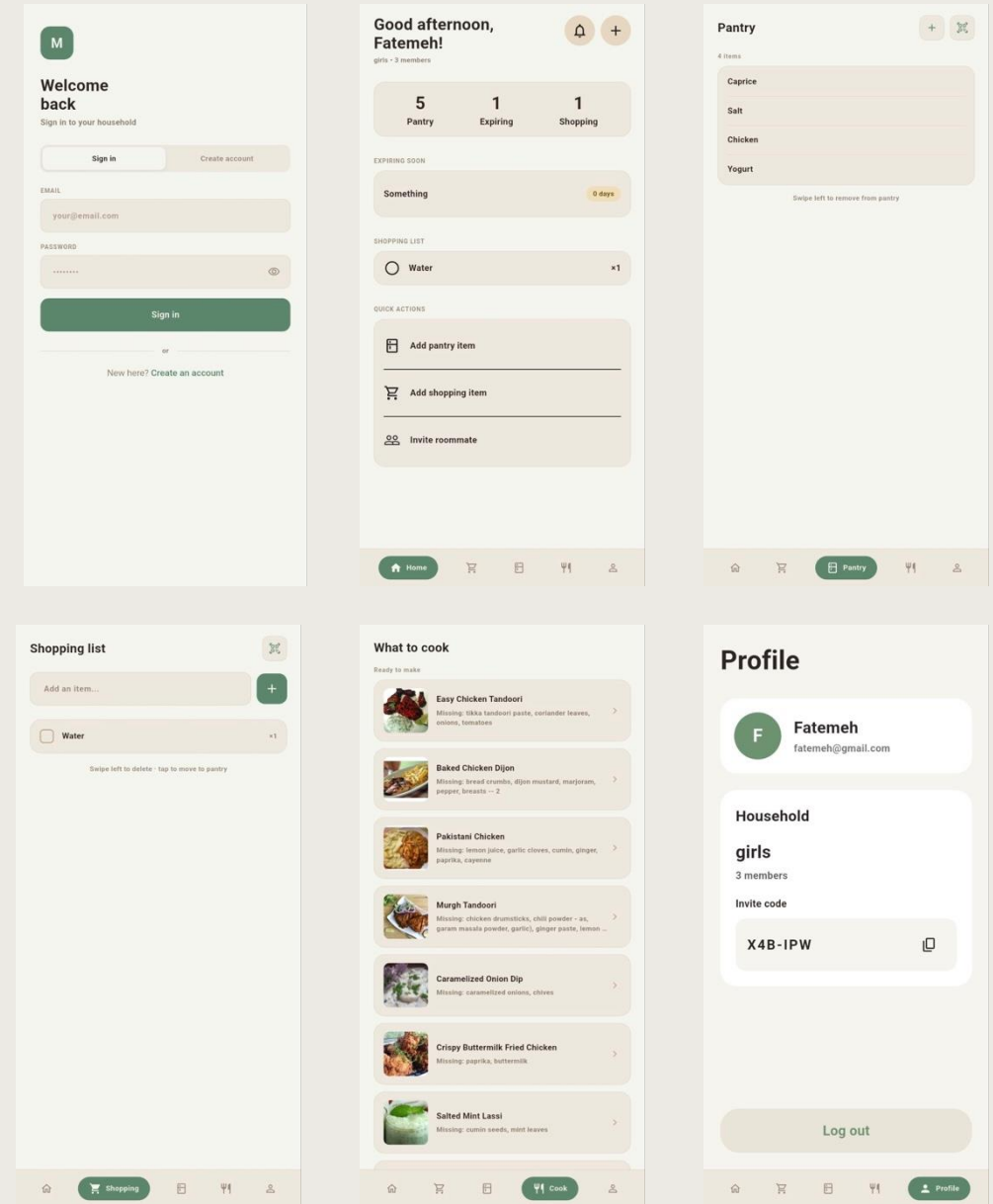


Barcode Scanning

- ▶ **Library:** mobile_scanner
 - ▶ **Flow:** User scans barcode → app calls OpenFoodFacts API to get product name
 - ▶ **Pre-fill:** Product name pre-fills the "Add Item" sheet — user confirms quantity and expiry
 - ▶ **Context:** Works for both pantry and shopping list (ScanContext enum)
 - ▶ **Fallback:** Manual search available as fallback
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The 5 Screens

- ▶ **Login / Household Setup:** sign in, create or join household via invite code
- ▶ **Home:** greeting, pantry overview, expiring items, shopping preview, quick actions
- ▶ **Pantry:** full item list, expiry pills, swipe to delete, scan or manual add
- ▶ **Cook Suggestions:** recipe cards ranked by pantry match, detail sheet with instructions
- ▶ **Shopping List:** shared list, check off moves item to pantry, scan to add



Who Built What

- **Fatemeh Hosseini:** app architecture, service locator, barcode scanning integration, session handling, Pantry and ShoppingBLoC
 - **Ana Teodora Cârjilă:** Firebase Auth backend, household create/join/invite system, home screen UI and quick actions
 - **Nour Al Houda Mahfood:** Drift database schema, CookBloc with Firestore sync, Cook screen with Spoonacular API integration
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Live Demo

- Scan an item
 - See pantry update
 - Watch recipe suggestions refresh
 - Check off shopping item
 - See it move to pantry
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Thank You

github.com/FaMaHo/mise
